

BLI: beliefs-about-future-beliefs

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Notation

day of logical inductor n

set of small beliefs S

level of discretisation $D = \{\frac{1}{d}; \frac{2}{d}; \dots; 1\}$

BLI $\mathbf{P}_n(\phi)$

LI $\mathbf{Q}_n(\phi)$

Constraints for BLI

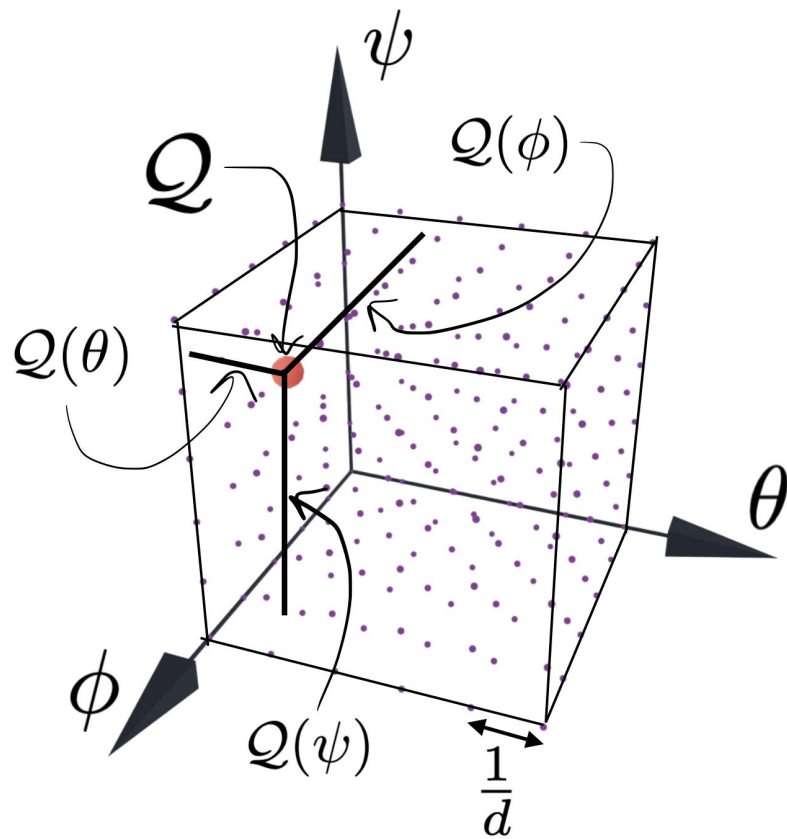
1. $\mathbf{P}_n(\phi) = \mathbf{Q}_n(\phi)$ where ϕ is a small belief
2. $\mathbf{P}_n(\phi|\mathbf{Q}_m = \mathcal{Q}) = \mathcal{Q}(\phi)$ where $n < m$ and $\mathcal{Q}$ is a written-out belief state which includes a probability for ϕ , and ϕ is a small belief
3. $\mathbf{P}_n(\phi|\mathbf{Q}_m = \mathcal{Q}_1 \wedge \mathbf{Q}_o = \mathcal{Q}_2) = \mathcal{Q}_2(\phi)$ where $n < m < o$ and ϕ is small
4. $\mathbf{P}_n(\mathbf{Q}_m = \mathcal{Q})$ must balance the equation

$$\mathbf{P}_n(\phi) = \sum_{\mathcal{Q} \in D^S} \mathbf{P}_n(\mathbf{Q}_{n+1} = \mathcal{Q}) \mathcal{Q}(\phi)$$

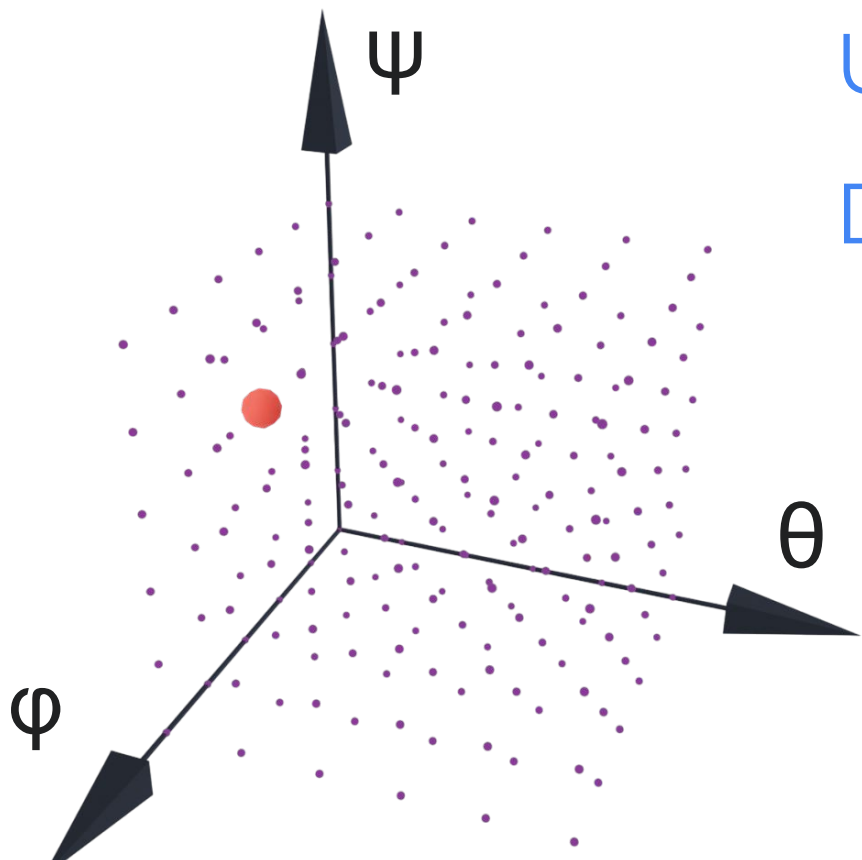
5. $\mathbf{P}_n(\mathbf{Q}_m = \mathcal{Q})$ must also balance the equation

$$\sum_{\mathcal{Q} \in D^S} \mathcal{P}_n(\mathbf{Q}_{n+1} = \mathcal{Q}) = 1$$

Future-belief as a point in lattice



Delta-superbelief



Update from 0

Division by 0

$$\mathbf{P}_n(\phi) = \sum_{\mathcal{Q} \in D^S} \mathbf{P}_n(\mathbf{Q}_{n+1} = \mathcal{Q}) \mathbf{P}_n(\phi | \mathbf{Q}_{n+1} = \mathcal{Q})$$

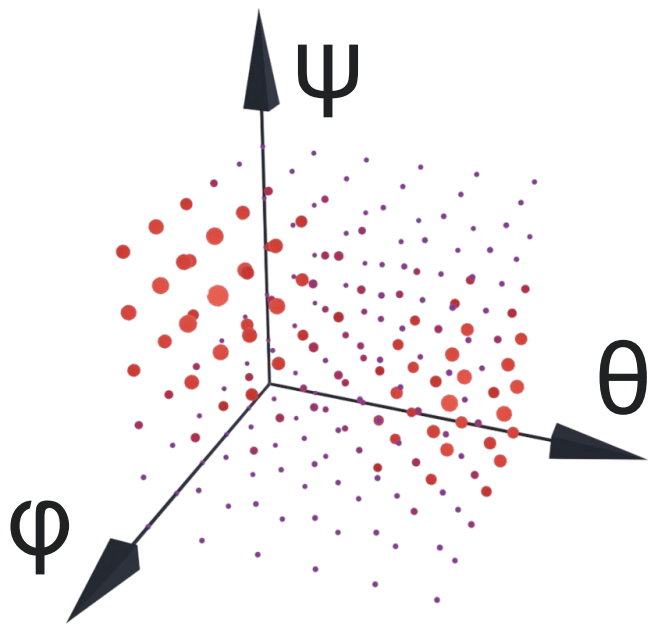
$$\mathbf{P}_n(\phi | \mathbf{Q}_{n+1} = \mathcal{Q}) = \frac{\mathbf{P}_n(\phi \wedge \mathbf{Q}_{n+1} = \mathcal{Q})}{\mathbf{P}_n(\mathbf{Q}_{n+1} = \mathcal{Q})}$$

$$\mathbf{P}_n(\mathbf{Q}_{n+1} = \mathcal{Q}) \neq 0$$

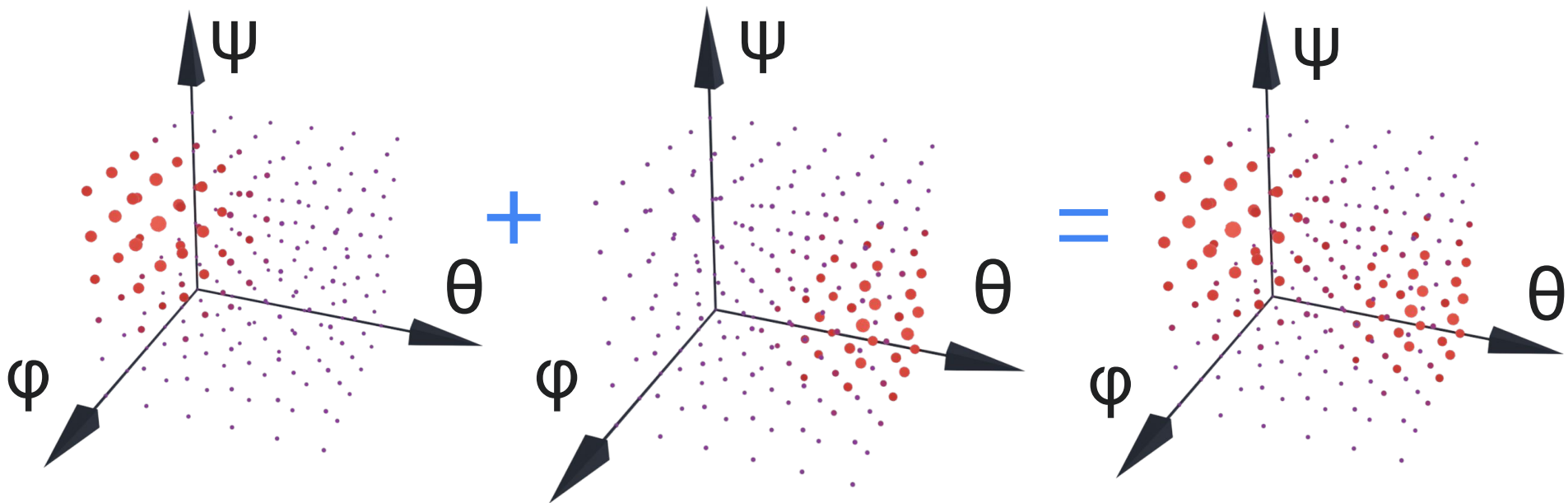
Superbelief as a vector

$$\vec{F} = (\mathbf{P}_n(\mathbf{Q}_{n+1} = \mathcal{Q}_0), \dots, \mathbf{P}_n(\mathbf{Q}_{n+1} = \mathcal{Q}_q))$$

$$q := |D^S| = d^s$$



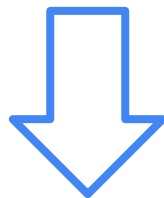
Linearity



Norm

$$\sum_{Q \in D^S} \mathbf{P}_n(\mathbf{Q}_{n+1} = Q) = 1$$

$$|\vec{F}| = \sum_{i=0}^{i=q} \vec{F}^i$$

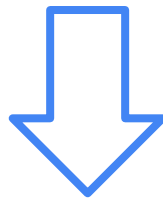


$$|\vec{F}| = 1$$

Einstein convolution

$$\mathbf{P}_n(\phi) = \sum_{\mathcal{Q} \in D^S} \mathbf{P}_n(\mathbf{Q}_{n+1} = \mathcal{Q}) \mathcal{Q}(\phi)$$

$$C[\vec{F}](\phi) := \sum_{i=0}^{i=q} \vec{F}^i \mathcal{Q}_i(\phi)$$



$$\mathbf{P}_n(\phi) = C[\vec{F}](\phi)$$

Linearity

$$|\alpha \vec{F}_1 + \beta \vec{F}_2| = \alpha |\vec{F}_1| + \beta |\vec{F}_2|$$

$$C[\alpha \vec{F}_1 + \beta \vec{F}_2](\phi) = \alpha C[\vec{F}_1](\phi) + \beta C[\vec{F}_2](\phi)$$

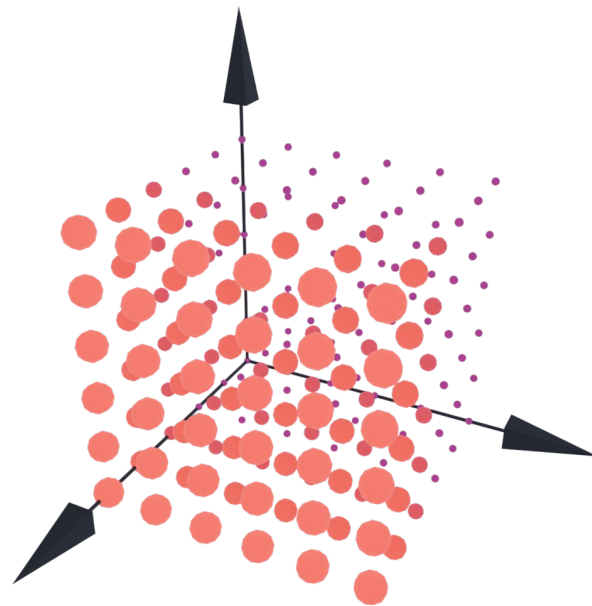
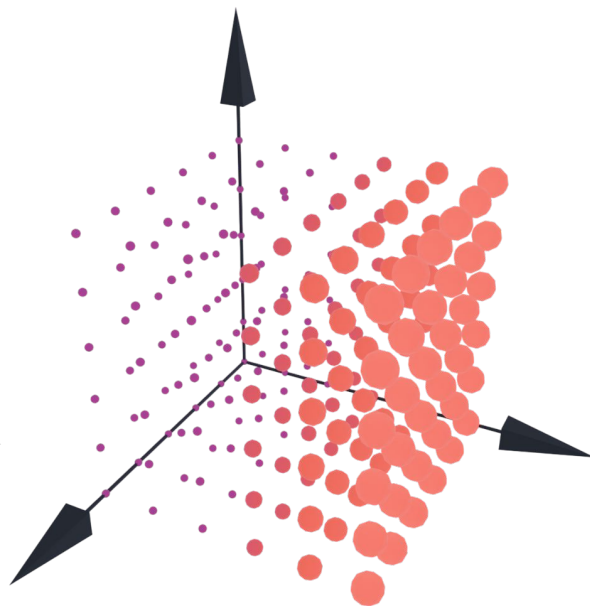
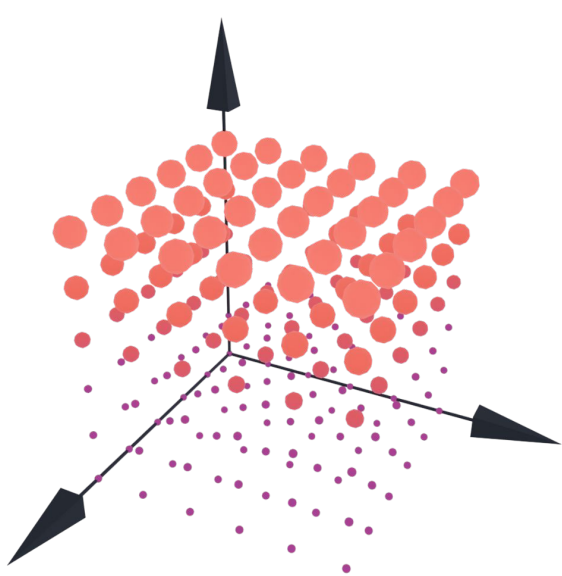
The basis

$$\mathbf{f} = (f_1, \dots, f_k)$$

$$\mathcal{B} = (\vec{F}_1, \dots, \vec{F}_k)$$

$$\vec{F} = (\mathbf{f}, \mathcal{B}) = \left(\begin{array}{c} f_1 \cdots f_k \end{array} \right) \left(\begin{array}{c} \vec{F}_1 \\ \vdots \\ \vec{F}_k \end{array} \right) = \sum_{i=1}^k f_i \vec{F}_i$$

The basis



Convolution matrix

$$C_{ij} := C[\vec{F}_i](\phi_j)$$

$$\mathbf{P}_n(\phi_j) = C[(\mathbf{f}, \mathcal{B})](\phi_j) = \sum_{i=1}^k f_i C[\vec{F}_i](\phi_j) = \sum_{i=1}^k f_i C_{ij}$$

The solution

$$\mathbf{f} = C_{ij}^{-1} \mathbf{P}_n(\phi_j)$$

$$\vec{F} = (\mathbf{f}, \mathcal{B})$$

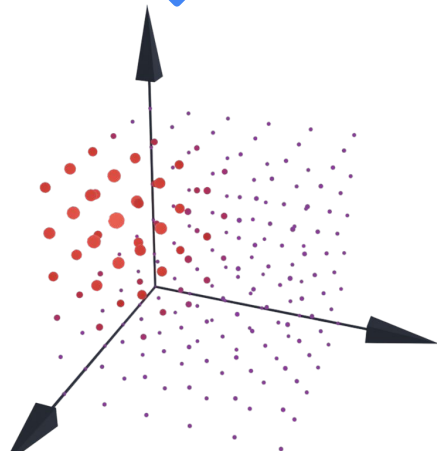
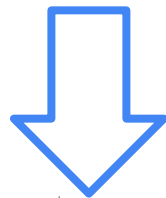
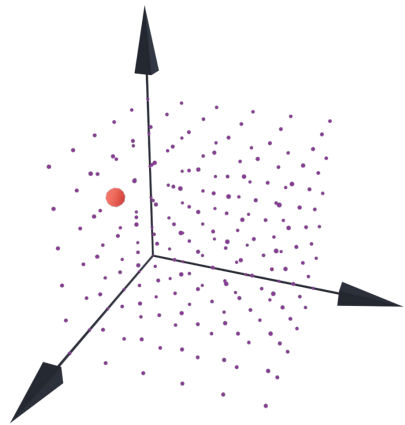
Max-entropy

$$\int_0^1 \rho(x) \ln \rho(x) dx \rightarrow \max$$

$$\int_0^1 \rho(x) dx = 1$$

$$\int_0^1 x \rho(x) dx = p$$

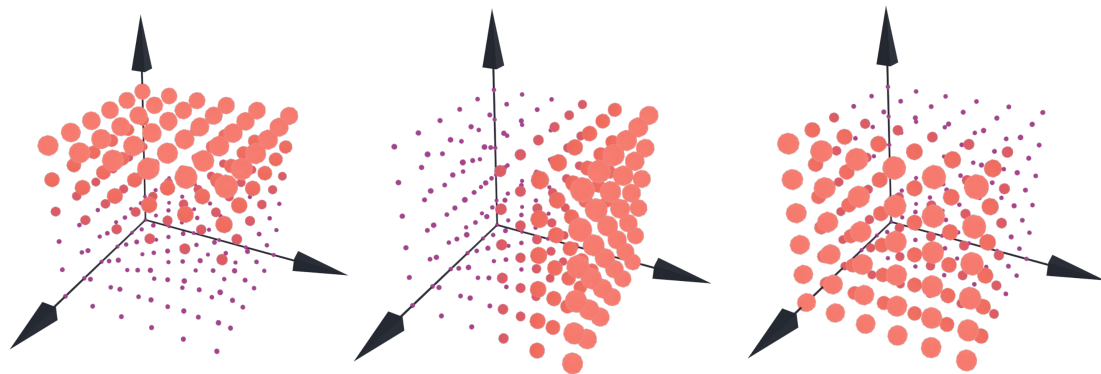
$$\rho(x) \sim e^{\gamma x}$$



The basis

$$F_i^j = \alpha e^{\gamma \mathcal{Q}_j(\phi_i)}$$

$$C_{ij} \sim \begin{bmatrix} 1 & \dots & e-1 \\ \vdots & \ddots & \vdots \\ e-1 & \dots & 1 \end{bmatrix}$$



Missing Details

Figuring out discretisation

Edge cases

Normalisation

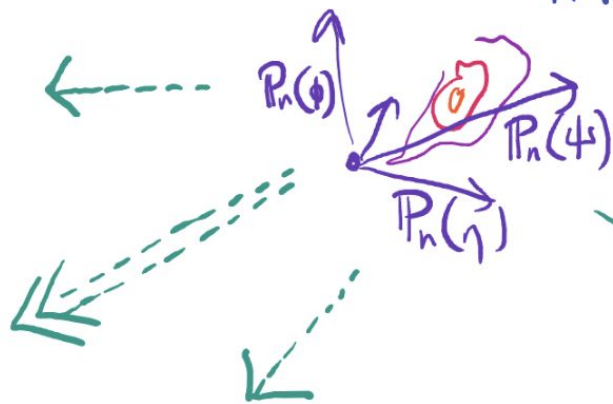
$[0; 1]$ boundaries

Small beliefs about future

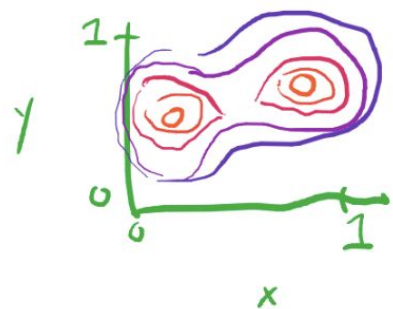
$$P_{n-1}(P_n(\phi)=x)$$



$P_{n-1}(Q_n=Q_n)$, our big
high-dimensional
distribution



$$P_{n-1}(P_n(\phi)=x \wedge P_n(\eta)=y)$$



$$P_{n-1}(P_n(\eta)=x)$$



$$P_{n-1}(P_n(\phi) > .2)$$

$$E_{n-1}\left(\frac{P_n(\phi)^2 + P_n(\psi)^3}{P(\eta)}\right) > .2$$

Thanks for attention!